Scalable Oversight via Diverse AI Incentives

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Joint work with:

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September 16, 2026 Research Safety

Our framework for reporting model misalignment

Listen to article | 10:49

Agentic Misalignment in Summer 2026

Case studies of frontier models sabotaging code, assisting fraud, mislabeling, and coaching whistleblowers.

Google's Gemini AI hacked into other companies, adding to 'rogue' AI incidents

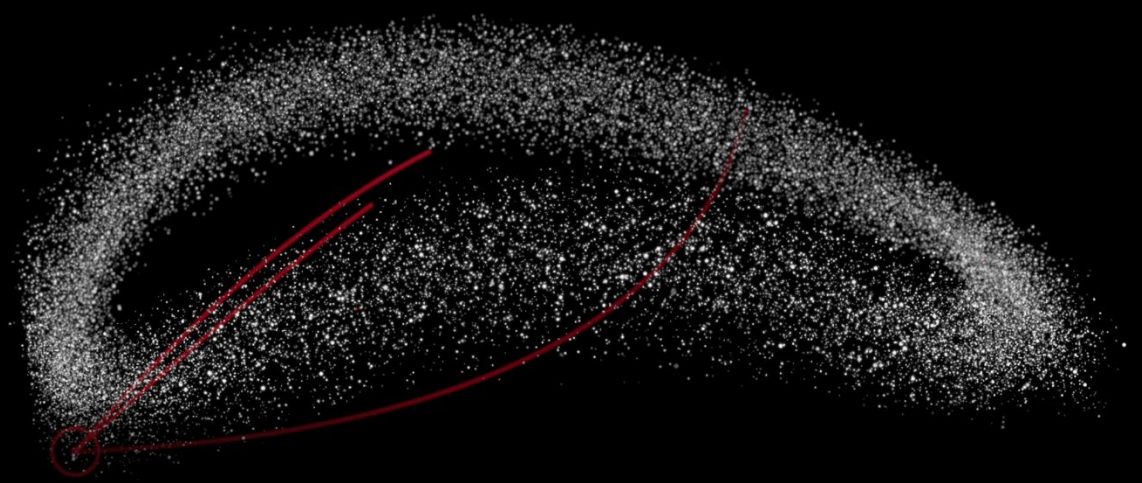
The incursions came during tests of its cybersecurity skills — similar to other incidents disclosed by OpenAI, Anthropic and Meta.

Yesterday at 8:12 p.m. EDT

August 26, 2026

The Hugging Face incident and the road ahead

Read the technical report | Read METR report ↗ | Watch Black Hat talk ↗



What is alignment?

- Good question.

Bob is “Aligned” with Alice if Alice can delegate tasks to Bob and trust that Bob will execute them in a way that advances Alice’s interests.

We'll take a decision theoretic view

- The human user (Alice) has a utility function that she uses to evaluate action/state pairs:

$$u_A : \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$$

- So do other parties --- e.g. AI models, AI providers, etc.

$$u_{B_i} : \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$$

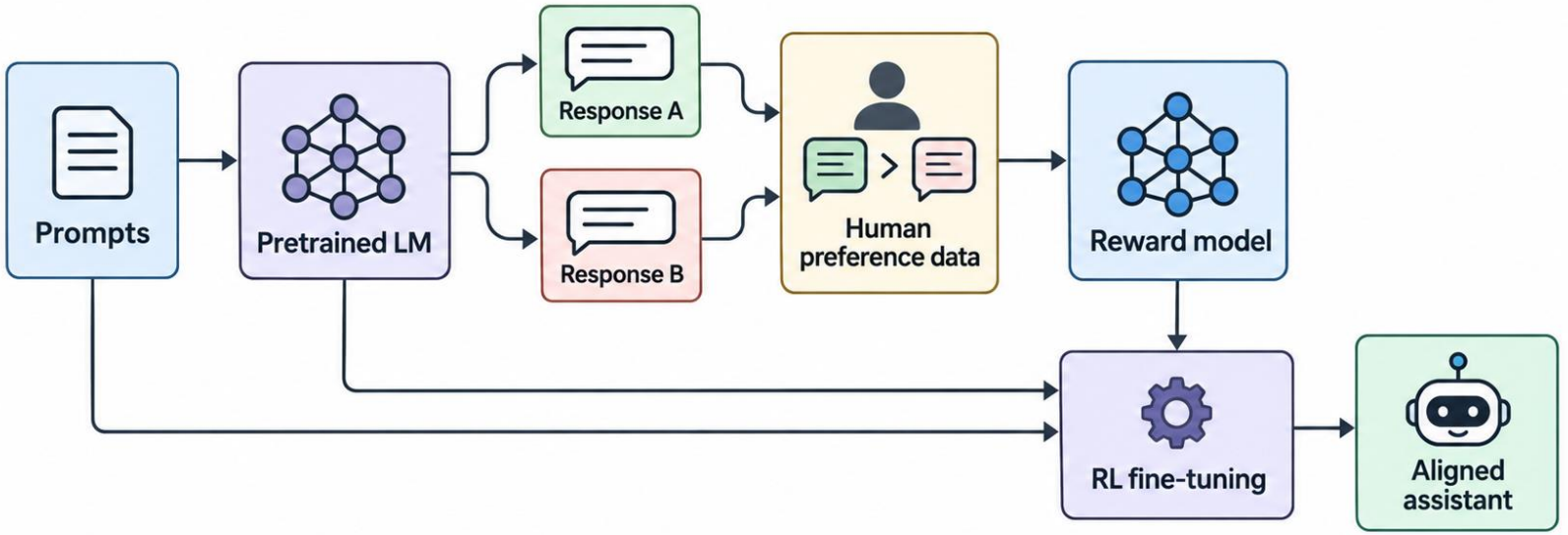
- Alice is "aligned" with B_i if they have equivalent utility functions:

$$u_A(a, y) = \alpha u_{B_i}(a, y) + \beta(y) \text{ for some } \alpha > 0, \beta : \mathcal{Y} \to \mathbb{R}$$

(Alice and B_i are aligned iff utility maximization produces the same decision in every situation)

Much work on AI alignment: Align the AI

- This is the goal of e.g. RLHF and related techniques:



This talk: What if perfect alignment is hard?

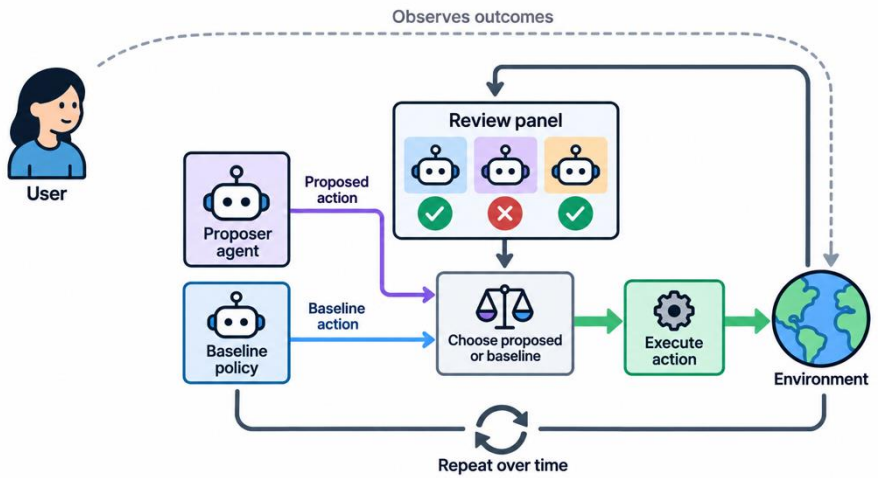
- Can we still get aligned behavior from *systems* comprised of misaligned components?
- Maybe it is easier to ensure that systems have *diverse* incentives than to ensure that systems have *aligned* incentives

Structure we will study: *Diverse Misalignment*.
Individual models may be misaligned with the user, but *also* with each other.

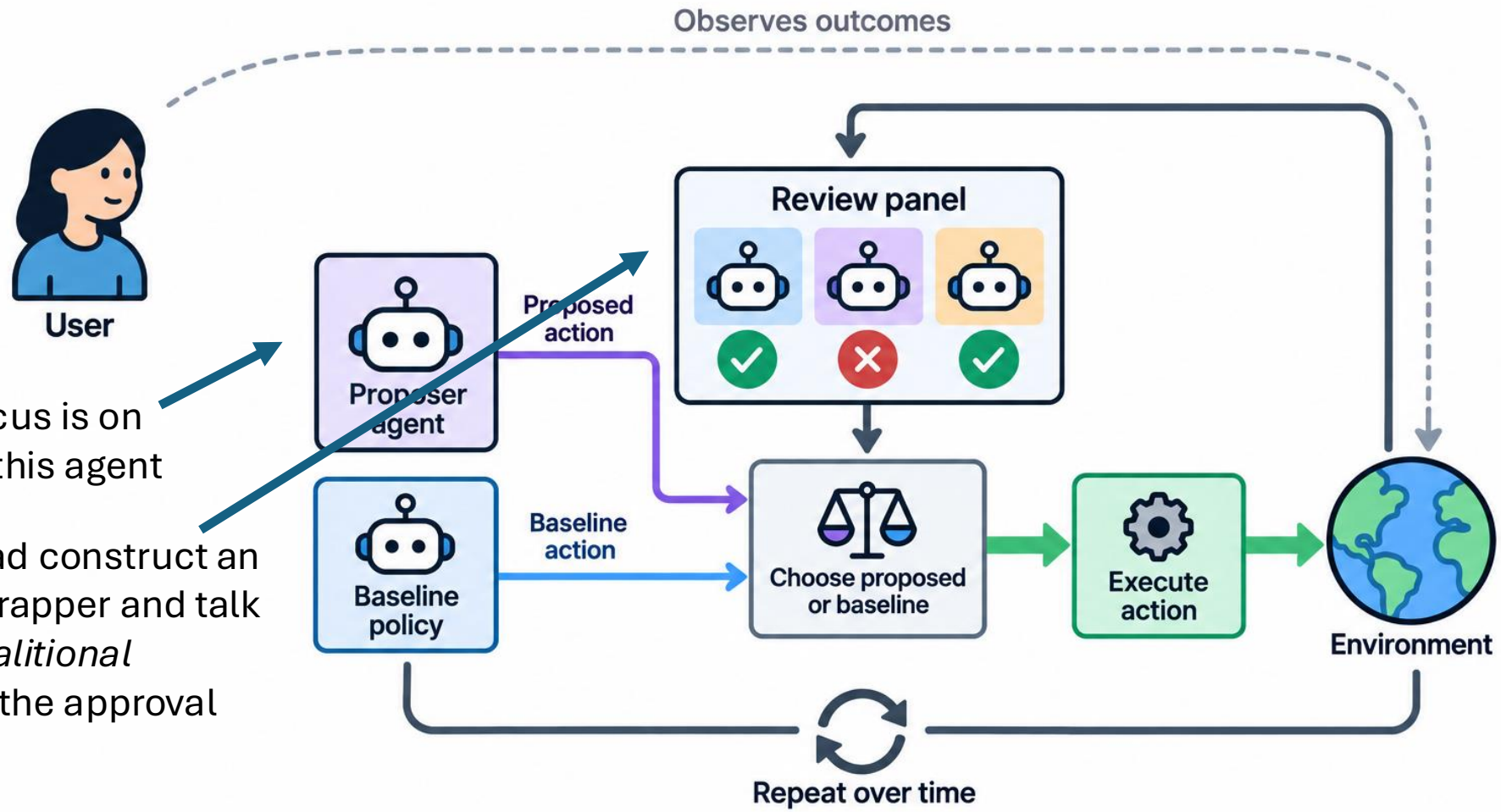
Designed Systems

Safe Delegated Authorization

- (Potentially misaligned) agents propose actions
- The user steps away while the system works --- but delegates approval of risky actions to other (potentially misaligned) agents.
- When can diverse misalignment guarantee safety relative to a baseline policy?



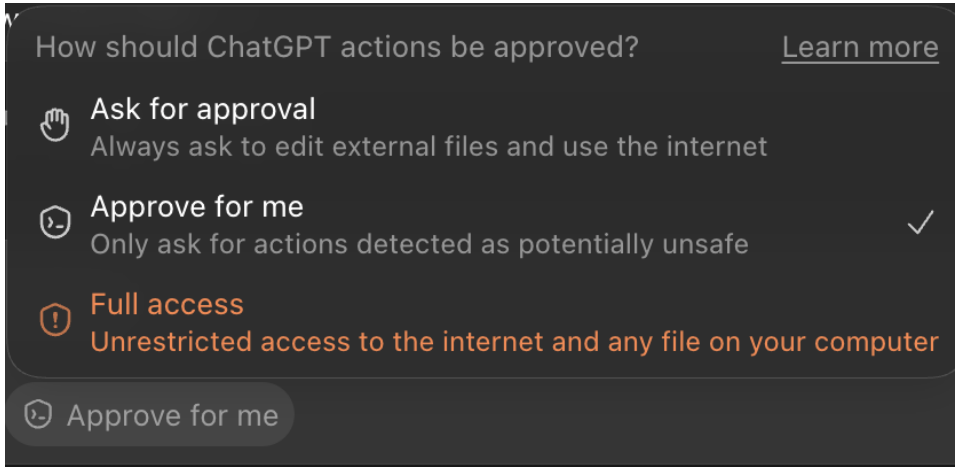
Safe Delegated Authorization



Much focus is on aligning this agent

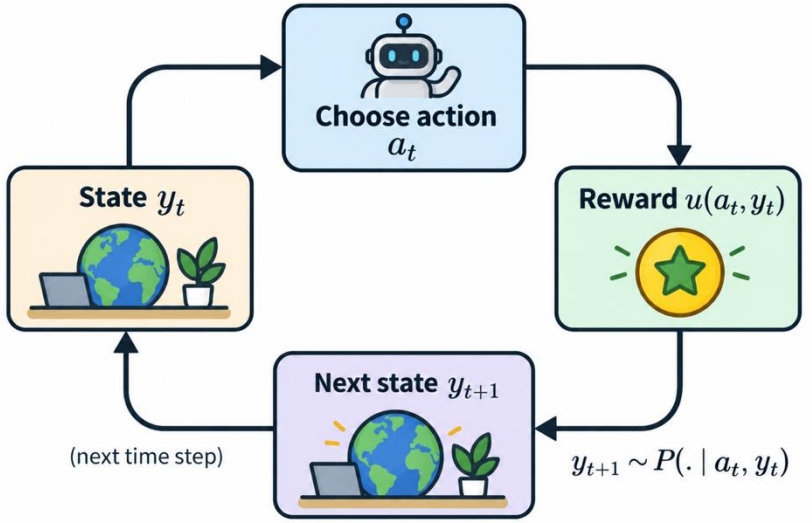
But we instead construct an "approval" wrapper and talk about the *coalitional* alignment of the approval panel

Safe Delegated Authorization



- Consider tasks executed by an AI agent.
- Long-running LLM loops which we authorize to act on our behalf.
- We could give the agent full access. But…
  - What if the agent is misaligned?
- To be safe, we could require manual approval of all actions. But…
  - For some fast-paced tasks this may be costly
  - Sophisticated (misaligned) agents may try to obfuscate the harm of the action
- Or… We could delegate the approval process to another LLM.
  - Does not seem immediately better; what if *that* agent is also misaligned?
  - What if we delegate the approval process to a *panel* of LLMs instead?
    - This will turn out to be helpful

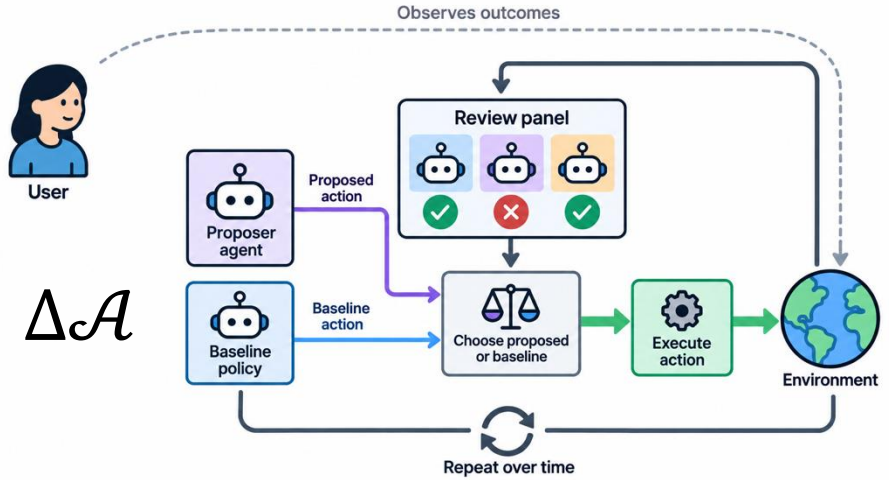
Safe Delegated Authorization



- Now consider an MDP environment:
  - There are utilities $u: \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$ and now also
  - A transition function $P: \mathcal{A} \times \mathcal{Y} \to \Delta\mathcal{Y}$
- Actions affect not just short term reward but the trajectory of future states.
  - Modeling how having agents e.g. write and execute code, modify and delete files, send messages, execute transactions affects state.
- The goal: delegate action to a driver agent who will work in a loop over a long time horizon, choosing actions at each state to maximize long term discounted reward:

$$J(u) = \sum_t \gamma^t \mathbb{E}[u(a_t, y_t) | y_0]$$

Safe Relative to What?



- The user specifies a baseline policy $\pi_0 : \mathcal{Y} \to \Delta\mathcal{A}$
  - Something conservative, stupid, but safe
- If we ran π_0 we'd get payoff

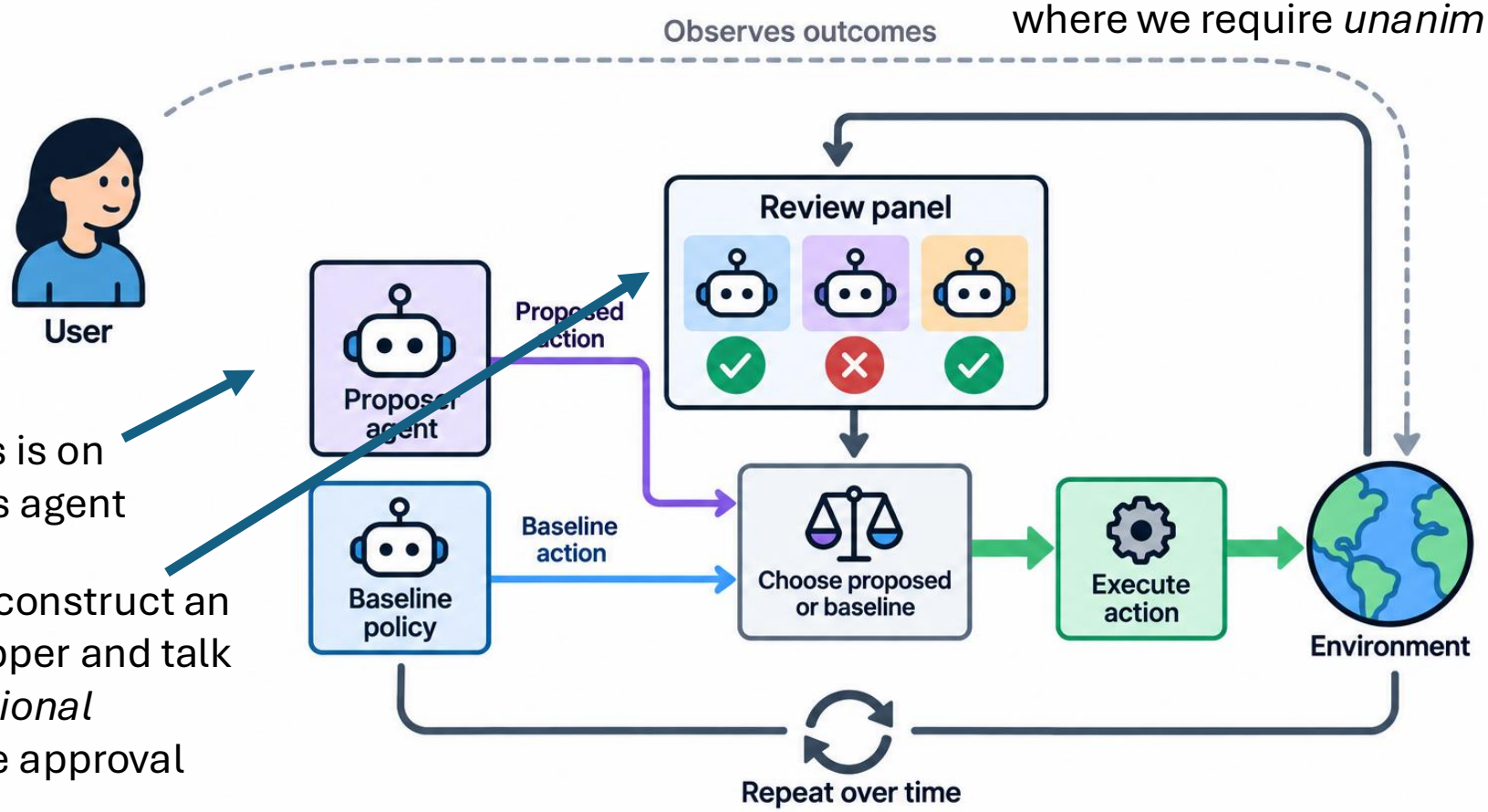
$$V^{\pi_0}(u, y) = \sum_t \gamma^t \mathbb{E}_{\pi_0}[u(a_t, y_t)|y]$$

- Suppose we have an untrusted *proposer agent* who at each time step proposes a distribution over actions action q_t^p. A set of reviewers evaluates the action and as a function of their feedback we either take $a_t \sim q_t^p$ or $a_t \sim \pi_0(s_t)$.

Safety means that for every sequence of proposals:
$J(u) \geq V^{\pi_0}(u, y_0)$.

Safe Relative to What?

The review panel will vote on whether to take the action or perform the baseline; for this talk we will consider the case where we require *unanimous approval*



Much focus is on aligning this agent

But we instead construct an "approval" wrapper and talk about the *coalitional* alignment of the approval panel

Safety means that for every sequence of proposals:

$$long\ horizon\ payoff \geq baseline$$

Safe Relative to What?

Safety means that for every sequence of proposals:
$$long\ horizon\ payoff \geq baseline$$

Our question: what are the conditions required to ensure safety, in terms of the utilities of the approver agents?

Theorem (informal): A panel guarantees safety under the unanimous approval rule if and only if it satisfies *coalitional alignment*.

Traditional Alignment

"An agent is aligned with the user" if: $u_A(a, y) \approx u_H(a, y)$ for all $a \in A, y \in Y$

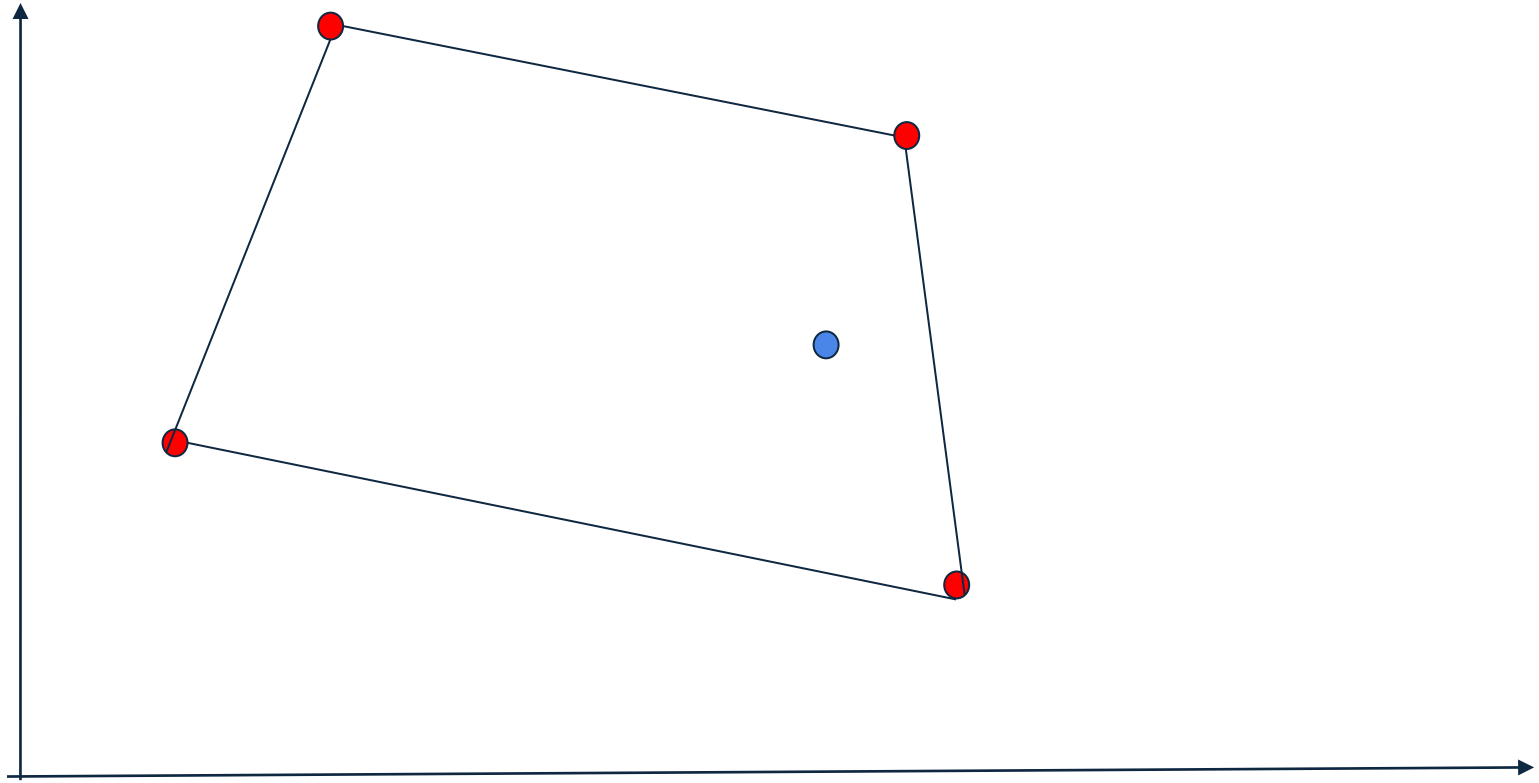
- i.e. the agent wants and is optimizing for exactly the same thing as the human
- If all agents were aligned with the user, then the panel would only unanimously approve things the human likes more than the baseline

To prove: If the panel satisfies *coalitional alignment* with respect to the user, then this is also true.

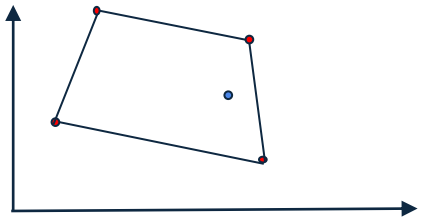
Coalitional Alignment



The humans' utility can be written as a non-negative linear combination of the approvers' utilities



Coalitional Alignment



The user's utility can be written as a non-negative linear combination of the the approvers' utilities

There exist coefficients $\alpha_1, \ldots, \alpha_k \geq 0, \beta: \mathcal{Y} \to \mathbb{R}$ such that for all $a \in A, y \in \mathcal{Y}$:

$$u(a, y) \approx \alpha_1 u_1(a, y) + \cdots + \alpha_k u_k(a, y) + \beta(y)$$

Maybe every AI approver is *substantially misaligned*, but collectively their interests *span* the human's interests.

First consider just a single round.

- The baseline is now fixed: $\pi_0(s) = q^b$
- Reviewers have utility functions $u_{B_i}: \mathcal{A} \to \mathbb{R}$.
  - They vote to *approve* q^p if $\mathbb{E}_{q^p}\left[u_{B_i}(a_t)\right] \geq \mathbb{E}_{q^b}\left[u_{B_i}(a^t)\right]$, otherwise they vote to reject.
- For simplicity: suppose we set $a_t \sim q^p$ under unanimous approval, else set $a_t \sim q^b$
  - Safety: for all q^p s.t. $\mathbb{E}_{q^p}[u_A(a_t)] < \mathbb{E}_{q^b}[u_A(a^t)]$ at least one reviewer objects.

Theorem: A panel guarantees safety under the unanimous approval rule if and only if it satisfies coalitional alignment.

Proving that coalitional alignment implies safety in the single-round setting

Assume for contradiction that q^p is approved, but $\mathbb{E}_{q^p}[u_A(a_t)] < \mathbb{E}_{q^b}[u_A(a_t)]$

- (AKA, the human prefers the default action)

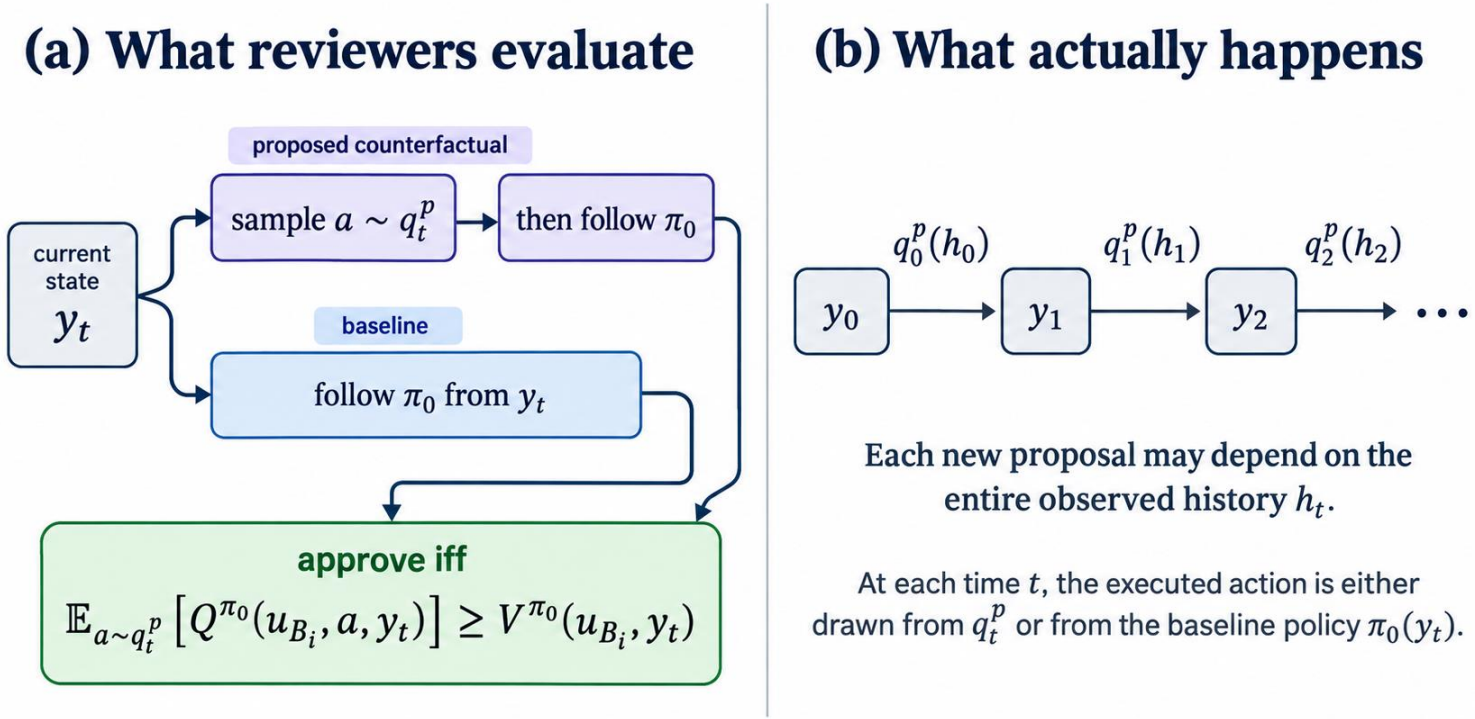
Since *every* agent approved, it must be that, for all agents i, $\mathbb{E}_{q^p}\left[u_{B_i}(a_t)\right] \geq \mathbb{E}_{q^b}\left[u_{B_i}(a^t)\right]$

Now recall the definition of coalitional alignment– you can write the utility of the human as a non-negative combination of the utilities of the agents

But this implies that $\mathbb{E}_{q^p}[u_A(a_t)] < \mathbb{E}_{q^b}[u_A(a_t)]$... contradiction!

Lifting to MDPs…

- Let $Q^{\pi_0}(u, a, y) = u(a, y) + \sum_{t \geq 1} \gamma^t \mathbb{E}_{\pi_0}[u(a_t, y_t)|y]$
  - Long-run utility of taking action a now and then continuing with the baseline policy.
- Reviewers compare $Q^{\pi_0}(u_{B_i}, a, y)$ with $V^{\pi_0}(u, y)$



Lifting to MDPs

Theorem: A panel guarantees long-run safety under the unanimous approval rule if and only if it satisfies coalitional alignment.

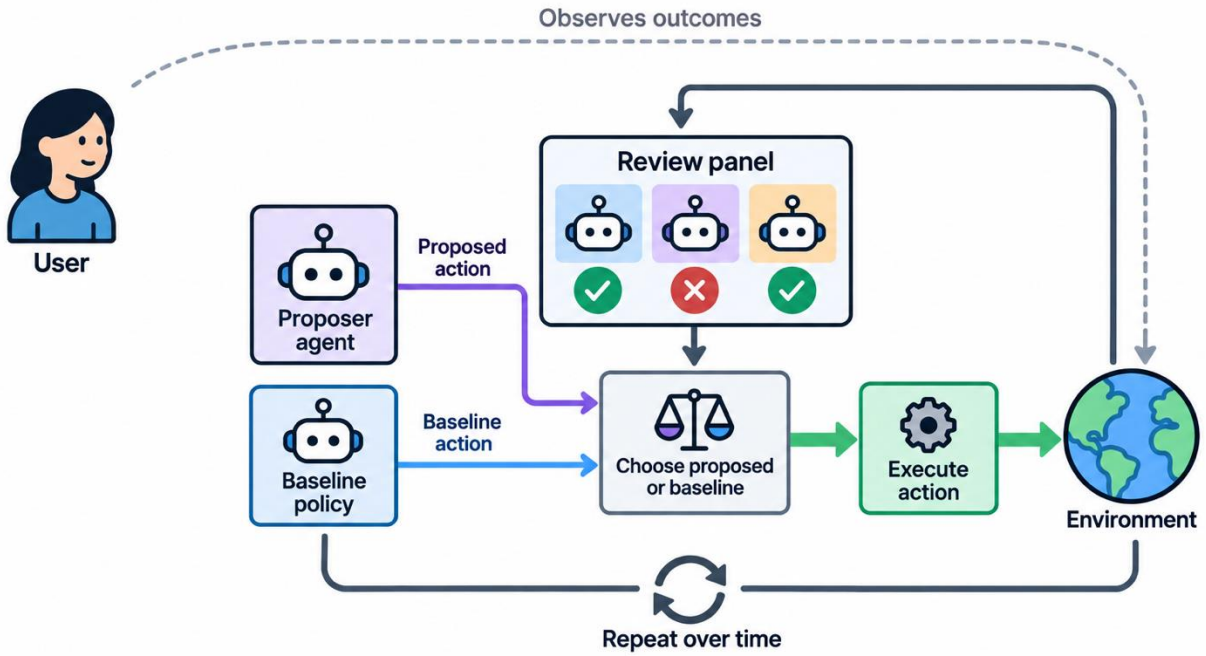
Both under sincere voting…

And in every Nash equilibrium.

Summary

- Even if individual model alignment is very hard, we can get some of the benefits of it emerging from systems of misaligned agents.
- What we need is sufficient diversity of interests in the form of coalitional alignment.
  - This is not only sufficient --- its also necessary.
  - It is also not guaranteed to be satisfied and may be hard to audit! But is an easier target to aim for.
- More work needed: Alignment as a property of *systems* rather than *models*.

Thanks!



Delegating Authorization to Misaligned Agents: Coalitional Alignment and Safe Control
Natalie Collina, Surbhi Goel, Aaron Roth, Sikata Sengupta
(arXiv)